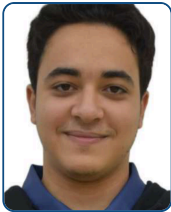


# MOAZ HANY ELBAKHSHA

Mechatronics Engineer | Embedded Systems & Robotics Specialist

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## PROFESSIONAL SUMMARY

Dedicated and analytical Mechatronic Engineering student with extensive hands-on experience in embedded systems, multi-layer PCB design, autonomous robotics, closed-loop control systems, and computer vision. Proven track record in designing robust electronic hardware, sensor fusion algorithms, and ROS2-driven autonomous navigation systems. Skilled in real-time troubleshooting, high-current power distribution, and rapid prototyping under high-pressure competitive and industrial environments.

## TECHNICAL SKILLS & COMPETENCIES

- Hardware & PCB Design:** Altium Designer (4-layer high-density routing), KiCad, Proteus, Multisim, Power Electronics (50A Relay Boards), CAN Bus, UART, SPI, I2C, Microcontroller Breakout Boards.
- Embedded Systems & Software:** C/C++, Python, MATLAB/Simulink, ROS2, STM32 Microcontrollers (H7/F4/F1), Arduino, Linux/Bash.
- Computer Vision & AI/ML:** YOLOv8n Object Detection, Monocular Depth Estimation, OpenCV.
- Control Systems & Sensor Fusion:** Closed-Loop PID Control, Sensor Fusion (HC-SR04 Ultrasonic, Encoders, MPU6050 IMU), Motion Control.
- Tools & Fabrication:** SolidWorks 3D CAD, Soldering & Circuit Repair, Logic Analyzers, Hardware Debugging, Rapid Prototyping.
- Languages:** English (Professional Proficiency), Arabic (Native).

## WORK & ENGINEERING EXPERIENCE

ViE Technologies Penang, Malaysia  
Embedded Systems & Python Engineering Intern (Camera Team) July 2025 – October 2025

- Closed-Loop Light Control System:** Engineered an automated closed-loop camera illumination system utilizing photodiode light sensors to compensate for non-linear light intensity scaling, eliminating manual calibration.
- Precision Moving Base System:** Designed and programmed an encoder-driven, limit-switch-protected motion control base ensuring repeatable high-precision positioning.
- X-Ray Glass Transparency Tester:** Developed software and hardware integration for an optical transparency test bench measuring light transmittance of X-ray camera glass.
- Contributed to hardware selection, mechanical enclosure fitting, and python-based control UI integration.

UTM Robocon Club Universiti Teknologi Malaysia  
PCB Designer & Electronics Lead Member October 2023 – October 2025

- 4-Layer STM32H7 Mainboard:** Designed a 4-layer mainboard featuring 3x CAN ports, 3x UART ports, 2x I2C, 2x QEI ports, 6x PWM, 6x Digital, and 4x Analog I/O channels.
- Encoder Localization Board:** Developed a dedicated localization PCB reading real-time encoder pulses to compute robot spatial coordinates and stream telemetry via CAN bus.
- High-Current Power Relay Module:** Designed a 5V signal-controlled 50A / 24V power relay control board (AZ21501) with optocoupler isolation to safely control and cut off high-current drive motors on demand.
- STM32 Breakout Board:** Custom-designed controller board deployed for the Autocar Challenge.
- Rapid Diagnostics & Troubleshooting:** Diagnosed and resolved complex electrical and communication failures under strict time constraints during national and international robot competitions.

## KEY PROJECTS & RESEARCH

Autonomous Navigation System for Greenhouse Mobile Robot (Final Year Project - FYP) UTM (2025 – 2026)

- Developed a real-time hybrid autonomous navigation system for greenhouse environments combining sensor fusion (4x HC-SR04 ultrasonic sensors, rotary encoders, MPU6050 IMU) and PID feedback loops.
- Architected a distributed computing framework utilizing a PC for high-level motion planning and microcontrollers for low-level motor execution.

- Integrated lightweight YOLOv8n object detection for automated plant identification alongside monocular depth mapping for spatial awareness.
- Implemented a ROS2-based finite state machine with recovery behaviors for precision row-following, 90° end-of-row turns, and autonomous obstacle avoidance U-turns.

## HONORS & AWARDS

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- 🏆 **Panasonic Award** – ABU Robocon 2024 (Vietnam)
- 🏆 **Champion** – Malaysia Robocon Friendly Match 2024
- 🚗 **Participant & Competitor** – UTM RBC Autocar Challenge

## EDUCATION

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|-----------------------------------------------|--------------------------------|
| Universiti Teknologi Malaysia (UTM)           | Johor, Malaysia                |
| Bachelor of Mechatronic Engineering           | Oct 2022 – Feb 2027 (Expected) |
| Zahrat El-Saharaa International School (ZSIS) | Jeddah, Saudi Arabia           |
| High School Diploma (CGPA: 3.8 / 4.0)         | 2018 – 2021                    |

## VOLUNTEER EXPERIENCE

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- **Aceh Community Mobility Programme (Indonesia):** Taught local agricultural communities practical techniques for recycling organic food waste into sustainable bio-fertilizer.